

MATH1231 CALCULUS

Chapter 4A

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4. Taylor Polynomials

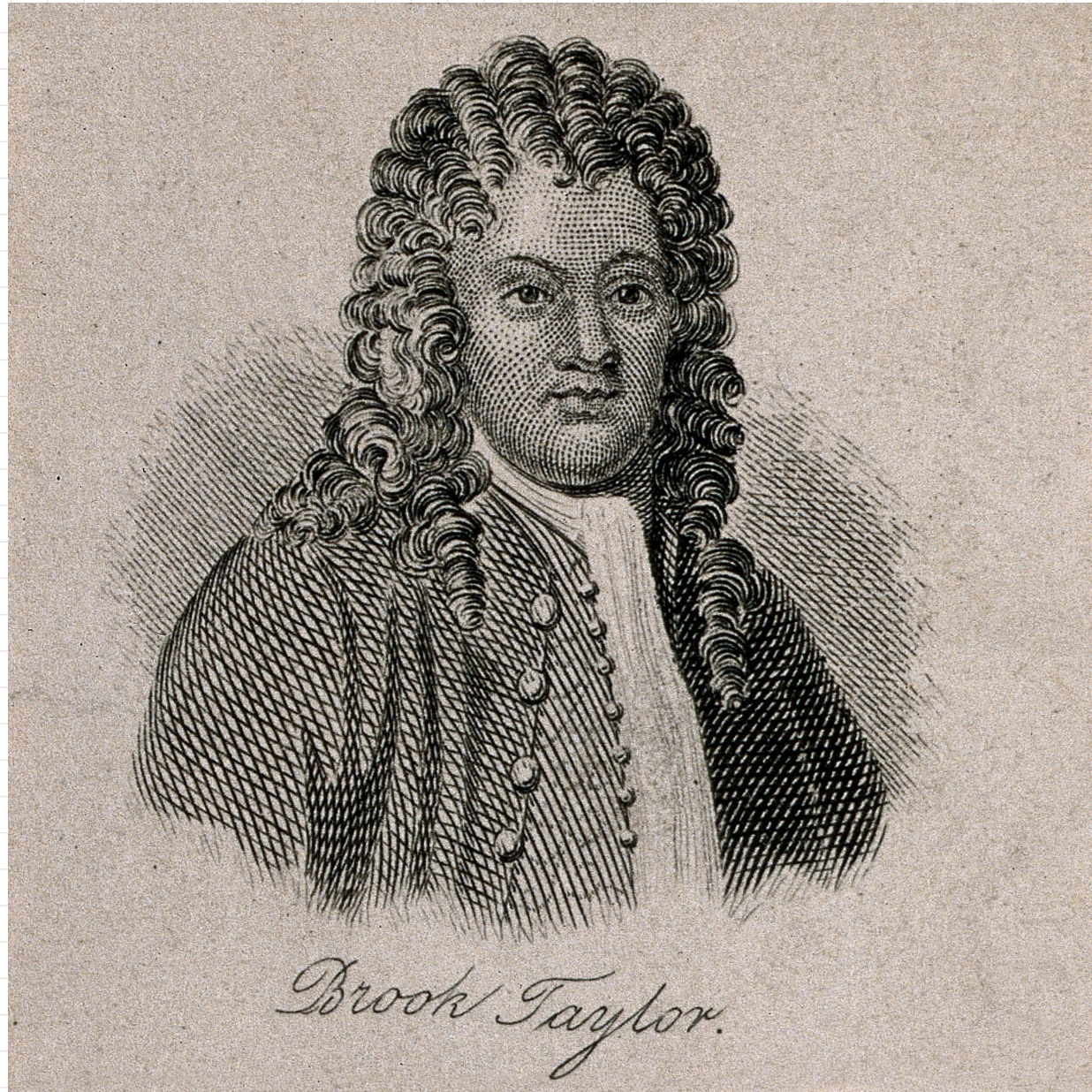


Figure: Brook Taylor 1685–1731

4.1) Taylor Polynomials

Differentiable functions such as $\sin$, $\cosh$, $\log$, $\exp$, ... can often be well approximated by polynomials

$$f(x) \approx a_0 + a_1x + a_2x^2 + \dots + a_Nx^N = p_N(x)$$

The higher the degree N the better the approximation, provided we make good choices for $a_0, a_1, \dots$

We will show that with ever more terms the polynomial converges to the function in many cases.

For example we will show that

$$e^{-x^2} = 1 - \frac{x^2}{1!} + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots$$

We can then integrate this function by integrating the simple powers of the RHS term by term.

Definition – Taylor Polynomial

Suppose that $f(x)$ is n times differentiable at $x = a$.

The n th Taylor polynomial of f about a is then given by

$$P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k$$

Note: $0! = 1$, by definition.

Example

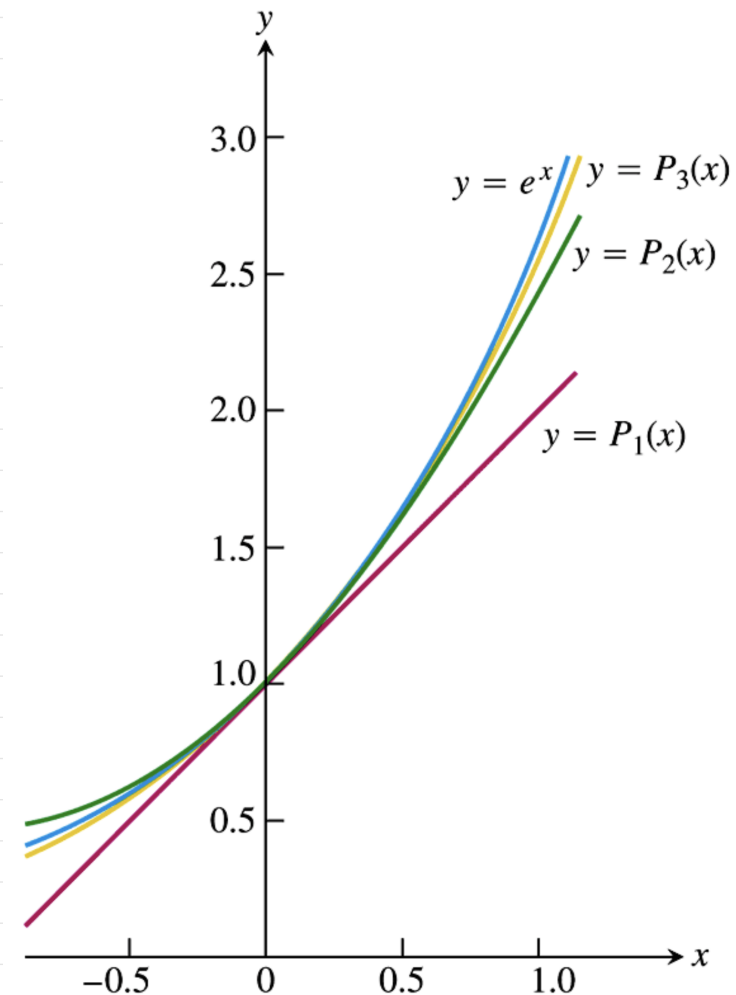
Ex: Compute the Taylor polynomial of order 3 generated by $f(x) = e^x$ at $x = 0 = a$.

We see:

$$\begin{aligned}f(x) &= e^x, & f(0) &= 1; \\f'(x) &= e^x, & f'(0) &= 1; \\f''(x) &= e^x, & f''(0) &= 1; \\f'''(x) &= e^x, & f'''(0) &= 1.\end{aligned}$$

Our Taylor polynomial definition gives

$$P_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}.$$



Note: The TP of degree n for $f(x)$ about $x = a$ shares the first n derivatives with $f(x)$ at $x = a$.

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{1}{2!}f''(a)(x-a)^2 + \frac{1}{3!}f^{(3)}(a)(x-a)^3 + \dots$$

Example

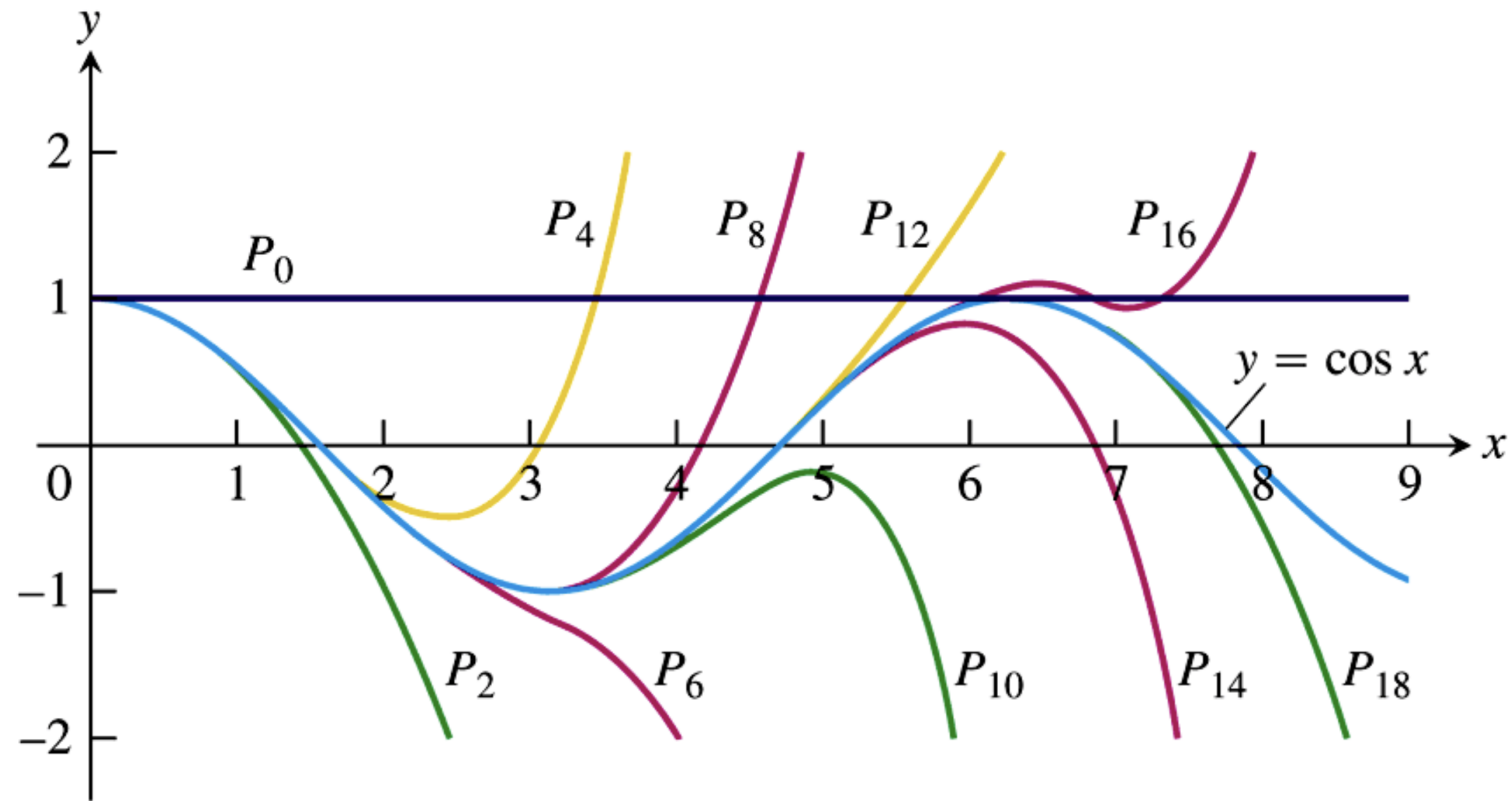
Find the Taylor polynomial of degree 7 for $f(x) = \sinh(x)$ about $x = 0$.

Example

Find the Taylor polynomial of degree 4 for $f(x) = \cos(x)$ about $x = \frac{\pi}{4}$.

Example

Find the Taylor polynomial of degree n for $f(x) = \cos(x)$ about $x = 0$.



Taylor polynomials provide good approximations to functions.

But how good?

What is the error?

Clearly it depends on the position x and the degree of the polynomial

(4.2.2) Taylor's Theorem & Lagrange Remainder Formula

If $f(x)$ has $n + 1$ continuous derivatives on an open interval I containing a then for each $x \in I$

$$f(x) = p_n(x) + R_{n+1}(x)$$

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k \quad \text{Taylor polynomial}$$

$$R_{n+1}(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - a)^{n+1} \quad \text{Lagrange remainder}$$

where c is some real number between a and x .

Connection with mean-value theorem

Taylor's theorem extends the mean value theorem (MVT).

$$\text{Taylor: } f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

For $n = 0$, we have $f(x) = f(a) + f'(c)(x-a)$

$$\Rightarrow f'(c) = \frac{f(x) - f(a)}{x - a} \quad \text{MVT}$$

Error estimations using the remainder term

If we approximate a function by a Taylor Polynomial then the **remainder** term provides the **error** in this approximation

$$|f(x) - p_n(x)| = |R_{n+1}(x)| \equiv |f^{(n+1)}(c)| \left| \frac{(x-a)^{n+1}}{(n+1)!} \right|$$

where c is between x and a

It is usually not possible to find c but it is often possible to find an **Upper Bound**, M for the error:

$$|f^{(n+1)}(c)| \left| \frac{(x-a)^{n+1}}{(n+1)!} \right| \leq M$$

Find the upper bound M by considering the maximum for all $x \in I$ and c between a and x .

Example

Let $f(x) = \cos x$. By considering the 2^{nd} Taylor polynomial for $f(x)$ about 0, estimate $\cos(\frac{1}{5})$ and find an upper bound for the error.

Example

Estimate the error of approximating $f(x) = e^x$ with the second-degree Taylor polynomial about $a = 0$ over the interval $(-0.5, 0.5)$.

Example

Find the minimum degree of the Taylor Polynomial for $f(x) = e^x$ about $a = 0$ to compute e to 5 decimal places.

(4.2.1) Classifying Stationary Points

Discuss the nature of the stationary points of (i) $f(x) = x^4$, and (ii) $g(x) = x^3$

Theorem

Suppose that f is n times differentiable at a and $f'(a) = 0$.
Then if $k \leq n$ and

$$f''(a) = f'''(a) = \dots = f^{(k-1)}(a) = 0 \quad \text{but} \quad f^{(k)}(a) \neq 0$$

we have

- (i) a **local minimum** at a if k is **even** and $f^{(k)}(a) > 0$
- (ii) a **local maximum** at a if k is **even** and $f^{(k)}(a) < 0$
- (iii) a **point of inflexion** at a if k is **odd**

Proof based on Taylor's Theorem – see notes

Exercise:

Use Taylor's Theorem to prove the following: Let f have continuous 1st- and 2nd-order derivatives on an interval I and suppose that $f'(a) = 0$ for some $a \in I$.

- (i) If $f'' < 0$ on I then f has a local maximum at a .
- (ii) If $f'' > 0$ on I then f has a local minimum at a .

If the degree of our Taylor “polynomial” becomes large (that is, $n \rightarrow \infty$) then we would like to determine if the error $R_{n+1}(x) \rightarrow 0$ for each x in the interval of interest. This would mean that a sum of an infinite number of powers of x indeed represents $f(x)$ accurately over the interval.

To understand precisely what we mean by $R_{n+1}(x) \rightarrow 0$ as $n \rightarrow \infty$, we will now examine the limits of **sequences**.